

Decoding of Postperformance Nonverbal Behavior of Tennis Players: Is a Shorter Response Time Associated With Higher Accuracy?

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Abstract

The decoding of nonverbal behavior is thought to be influenced by both bottom-up automatic perceptual and top-down higher order cognitive processes. In this study, we aimed to provide initial evidence for the association between response time and accuracy, as a first step toward understanding how such processes may operate. To that end, 145 individuals (66 female; $M_{age} = 32.18$) were presented 80 videos of amateur tennis players' nonverbal behavior directly after the rally. After each video, the participants were asked to estimate whether the previous point was won or lost. The results showed that the response time was about a half second faster for correct ($M = 3,052.78$ ms, $SD = 840$) than for incorrect ($M = 3,567.77$ ms, $SD = 1,048.15$) responses. These findings provide first evidence that faster responses are related to greater accuracy, suggesting that efficient, possibly more intuitive processes may facilitate decoding in this context.

Keywords: affect, emotions, automaticity, appraisal, negativity bias

The belief that athletes' nonverbal behavior may provide insights into their underlying internal states has led to an increasing interest in studying both their encoding and decoding processes (Furley & Schweizer, 2020). Encoding processes refer to the way individuals express internal states through nonverbal means, while decoding processes refer to how other individuals make sense of these nonverbal cues (Furley, 2021). Considering the effects athletes' nonverbal behavior can have on those around them (Fritsch, Redlich, et al., 2022), a better understanding of the underlying psychological processes is warranted from both a theoretical and an applied perspective (Furley, 2021).

In the present study, the focus is on tennis players' nonverbal behavior immediately following

the end of a rally, commonly referred to as post-performance nonverbal behavior (Fritsch, 2024). For this kind of behavior, it seems useful to draw on appraisal approaches of emotions, which posit that affective processes emerge from an individual's interpretation of a personally relevant situation (e.g., Scherer, 2013). This theoretical assumption can account for both inter- and intraindividual differences in affective responses to similar situations (Scherer & Moors, 2019). While appraisal approaches of emotions highlight the idiosyncratic nature of appraisal processes, meta-analytical evidence shows that situations appraised as inconducive to one's goals are typically linked to negative affective states, whereas goal-conducive situations are typically linked to positive affective states (Yeo & Ong, 2024). Applied to the end of rally in tennis, a won point can be appraised as goal conducive, thereby eliciting positive affective responses, whereas a

lost point is appraised as goal obstructive, eliciting negative affective responses (Jekauc et al., 2021). Thus, while it is important to acknowledge that postperformance nonverbal behavior may also convey information about other psychological states than affect (Fridlund, 1994), these appraisal processes suggest that post-performance nonverbal is strongly associated with affective processes (Fritsch, 2024). This association may, in turn, influence the decoding processes of postperformance nonverbal behavior (Furley, 2021).

In the present study, we specifically focus on the response time as a behavioral indicator of the underlying decoding processes of postperformance nonverbal behavior, as this is a commonly used measure in cognitive psychology to distinguish more automatic and controlled forms of information processing (Logan, 1988; Schneider & Shiffrin, 1977). While appraisal approaches of emotions suggest an inherent link between athletes' postperformance nonverbal behavior and affective processes (Jekauc et al., 2021), the biocultural framework of nonverbal behavior (Furley, 2021) offers a suitable perspective for understanding the decoding mechanisms. This framework posits that both bottom-up automatic perceptual and higher order top-down cognitive processes influence the decoding of nonverbal behavior (Furley, 2021). This reasoning also aligns with the dynamic interactive theory of personal construal, which emphasizes how the interaction between lower level sensory perception and higher order social cognition processes dynamically shape the perception of other's behavior (Freeman & Ambady, 2011).

Given the fundamental role that nonverbal behavior plays in social interactions, enhancing the understanding of its decoding processes is crucial not only from a theoretical perspective but also from an applied one. Considering the impact that observing an opponent's or teammate's specific behavior can have on an individual (for a review, see Furley, 2021), athletes might adopt strategies that specifically target either automatic perceptual or higher order cognitive processes. For instance, if the rapid, relatively automatic decoding of an opponent's nonverbal behavior (e.g., a

dominant gesture) undermines an athlete's self-confidence, attentional control strategies—such as deliberately redirecting focus toward task-relevant cues (e.g., the ball or tactical positioning)—may help to reduce the influence of these automatic perceptual processes. Conversely, in situations where the nonverbal behavior is more ambiguous, athletes may benefit from strategies that engage higher order cognitive processes, such as reappraising the meaning of an opponent's behavior in light of contextual information.

In line with the biocultural framework of nonverbal behavior, it has been proposed that the relative influence of automatic perceptual and higher order cognitive processes depends on the type of nonverbal behavior (Furley, 2021). In this regard, response time is considered a reliable behavioral indicator to provide indirect insights into decoding processes, with faster responses sometimes interpreted as reflecting more automatic perceptual processing (e.g., Cardoso et al., 2021; Logan, 1988; Schneider & Shiffrin, 1977). This rationale is consistent with findings that accuracy tends to be high and response times fast in perceptual judgment tasks with strong stimulus signals (Palmer et al., 2005). Hence, the investigation of response times in association with response accuracy offers potential to provide the first empirical test of important tenets of biocultural theorizing of nonverbal behavior in sports (Furley, 2021).

There is growing evidence on nonverbal behavior in sports showing that athletes reveal information through their nonverbal behavior, which others can detect with a relatively high degree of accuracy (for a review, see Furley, 2021). Focusing on the post-performance nonverbal behavior, one line of studies has shown short video excerpts of tennis players' nonverbal behavior immediately after the end of rally to participants and asked them to estimate whether the player just won or lost a point (Fritsch, Preine, Jekauc, 2022; Fritsch et al., 2023). The results have shown, among other findings, that lost points are easier to recognize than won points. This finding aligns with the negativity bias, a well-established psychological phenomenon that holds that events harmful to one's well-being and/or goals

have a stronger impact on mental states than events that are beneficial (Baumeister et al., 2001; Rozin & Royzman, 2001).

The Present Research

Previous research has primarily focused on how variations in the stimulus material (e.g., winning vs. losing situations) influence recognition accuracy of postperformance nonverbal behavior (e.g., Fritsch, Preine, Jekauc, 2022). Notably, recognition accuracy depends not only on how athletes encode nonverbal behavior but also on how observers decode it (Furley, 2021). Theoretical accounts suggest that decoding is shaped by both bottom-up automatic perceptual processes and higher order top-down cognitive processes (Freeman & Ambady, 2011; Furley, 2021), which, however, has not yet been systematically examined yet. For this reason, the present study focuses on response time as a behavioral indicator that may provide insights into the decoding of nonverbal behavior, based on the theoretical considerations outlined above (Furley, 2021). Faster response times may, under certain conditions, reflect more automatic perceptual processes (Cardoso et al., 2021; Logan, 1988; Schneider & Shiffrin, 1977), consistent with theoretical assumptions in the literature (e.g., Furley, 2021; Zebrowitz & Collins, 1997). Based on this reasoning, we hypothesized that response times would be shorter for correct than for incorrect responses when participants inferred whether a point was won or lost after brief observations of a player's nonverbal behavior. In addition, we exploratively examined whether the association between response time and accuracy might vary as a function of situation type (i.e., point won vs. lost) or participant gender.

Methods

Participants

Apart from participants having to be at least 18 years of age, there were no inclusion criteria for the study. The power analysis was conducted with G*Power 3.1 for a paired-sample *t* test. Conservatively assuming a small effect of $d = 0.25$, power of 0.80, and an alpha error of .05, the

calculation resulted in a sample of at least 101 participants. Individuals were directly approached through personal contacts and invited to participate in the study. In addition, they were recruited following the snowball sampling principle, meaning that initial participants were asked to recommend further individuals from their personal networks who might also be willing to take part. There was no reward for the participation in the study. A total of 145 individuals (66 female) with a mean age of 32.18 (standard deviation [SD] = 13.94) years agreed to participate in the study. Ethical approval was granted by the ethic's committee of Karlsruhe Institute of Technology. The video filming and editing as well as the recruiting of the participants took place at the Karlsruhe Institute of Technology, where the first author of the article was employed at the time of data collection.

Video Excerpts

Amateur tennis matches were filmed for the purpose of the present study. The video excerpts were based on 18 amateur tennis players (six females and 12 males) who were filmed during a competitive tennis match (i.e., either of the ongoing season or a competitive tournament). For the filming, the camera was positioned on a tripod close to the net, filming the players from the front including the whole body. Subsequently, the footage was cut in such a way that it included the first 3 s after the end of a rally. Criteria for points being used in the study were: (a) the trajectory of the ball did not allow to make conclusions about the point outcome and (b) players were to be seen from the front on the video excerpts. This selection process resulted in a total of 80 points with 40 lost and 40 won points with each player being shown on average on 4.44 videos.

Procedure

We used the free software OpenSesame to present the video excerpts to the participants. To ensure good visibility of the players on the videos, the study was conducted on a computer or laptop in a quiet environment. The participants were asked to have a distance of about 60 cm to the

device showing the videos. At the beginning of the study, standardized on-screen instructions were presented to the participants. These instructions included the information that the participation is voluntary, that participants could withdraw at any point without giving any justification, and that their data would be analyzed anonymously. The participants gave their consent by confirming their participation in the software. Two test video excerpts including a won and a lost point were shown to the participants to illustrate the following procedure. Thereafter, the 80 video excerpts were shown to the participants in a random order. Each video excerpt was shown only once and without sound. After viewing the players' nonverbal behavior, the participants were asked to estimate whether the shown player just won or lost a point. The answer was given by clicking with the cursor on response option "won point" or "lost point" in the software. This procedure follows previous studies in which participants were able to distinguish between won and lost points at above-chance levels based on tennis players' nonverbal behavior (e.g., [Fritsch, Preine, Jekauc, 2022](#); [Fritsch et al., 2023](#)). To assess the response time, the software tracked how long the participants needed to give a response after the video was shown. Notably, participants were asked to answer as accurately as possible, while the speed of their response was not emphasized.

The reason for this was that we wanted to assess their natural response time when they tried to make meaning of the visible nonverbal behavior. Moreover, the participants were not given any feedback for their responses during the entire procedure, but, if wanted, they could give the information about the amount of correct responses at the end of the study.

Measurement

Several indicators were created for the three independent variables used in the present study. First, regarding the importance of the situation, as illustrated in Table 1, we created two indicators for the proximity to the end of a match section (i.e., remaining games, remaining points), and two indicators for the closeness of the score (i.e., game difference, point difference). Second,

regarding point outcome, a dichotomous variable was used to distinguish whether the player shown on the video excerpt had won or lost a point. A situation with a lost point was coded "0" and a situation with a won point was coded "1." Third, concerning tennis expertise, a dichotomous variable was created to distinguish whether or not a participant had been a member of a tennis club. When the participant had not been a member of a tennis club the situation was coded "0" and when he/she had been a member of a tennis club it was coded "1." Finally, based on the answers of the participants, the dependent variable was a dichotomous variable with the two outcomes "the estimate of point outcome was correct" coded "1" or "the estimate of point outcome was not correct" coded "0."

Statistical Analyses

To assess the association between the response time and the accuracy of their responses, in line with the recommendations by Berger and Kiefer (2021), we first excluded answers with a response time above or below two SDs (2.7%). Based on the remaining trials, we computed for each participant (a) the average response time for correctly estimated video excerpts and (b) the average response time for incorrectly estimated video excerpts. These differences were assessed with a paired-sample *t* test. In addition, we exploratorily examined whether the effect of response accuracy on response time was moderated by (a) the type of situation (i.e., won vs. lost rallies) or (b) the gender of participants (i.e., male vs. female). To this end, we conducted a mixed $2 \times 2 \times 2$ analysis of variance with the two within-subject factors response accuracy and type of situation, as well as the between-subject factor gender. The significance level was set a priori at 5% for all analyses.

Results

The average rates of correct responses in total were 69.72% (SD=6.97; 95% CI [68.60%, 70.84%]), 65.64% (SD=11.97; 95% CI [63.68%, 67.60%]) for won points, and 73.81% (SD = 10.74; 95% CI [72.06%, 75.56%]) for lost points.

Table 1
Descriptive statistics for response times

	<i>M</i> in milliseconds	<i>CI</i> (95%)
All points		
All responses	3200.03 (<i>SD</i> = 877.80)	3056.66 – 3343.40
Correct responses	3052.78 (<i>SD</i> = 840.00)	2915.58 – 3189.98
Incorrect responses	3567.77 (<i>SD</i> = 1048.15)	3396.57 – 3738.97
Won points correct responses		
Female participants	3302.15 (<i>SD</i> = 996.32)	3059.94 – 3544.63
Male participants	2956.49 (<i>SD</i> = 774.08)	2784.70 – 3128.28
All participants	3113.82 (<i>SD</i> = 895.81)	2967.50 – 3260.14
Lost points correct responses		
Female participants	3103.18 (<i>SD</i> = 924.49)	2348.77 – 3857.59
Male participants	2901.57 (<i>SD</i> = 767.51)	2731.24 – 3071.90
All participants	2993.34 (<i>SD</i> = 845.59)	2855.23 – 3131.45
Won points incorrect responses		
Female participants	3634.69 (<i>SD</i> = 1291.25)	3320.77 – 3948.60
Male participants	3445.73 (<i>SD</i> = 950.98)	3234.68 – 3656.77
All participants	3531.74 (<i>SD</i> = 1118.65)	3349.03 – 3714.45
Lost points incorrect responses		
Female participants	3750.56 (<i>SD</i> = 1354.73)	3421.21 – 4079.91
Male participants	3582.08 (<i>SD</i> = 1081.89)	3341.98 – 3822.18
All participants	3658.77 (<i>SD</i> = 1212.24)	3460.77 – 3856.77

The Association Between Response Time and Recognition Accuracy

The average response time for all points is shown in Table 1, separated also for won and lost points, as well as male and female participants. The comparison between correct (*M* = 3,052.78 ms, *SD* = 840.00) and incorrect answers (*M* =

3,567.77 ms, *SD* = 1,048.15) showed that the response time was significantly higher for incorrect responses than for correct responses, $t(144) = 11.66$, $p < .01$, $d = 0.96$.

Explorative Analyses

The full results of the mixed $2 \times 2 \times 2$ analysis of variance are shown in Table 2. With relevance

for our research question, the analysis revealed a significant interaction between type of situation and response accuracy, $F(1, 143) = 7.23$, $p = .008$, $\eta^2_p = .05$. As shown in Figure 1, response times were faster for correct than for incorrect responses in both situations, with this difference being more pronounced when points were lost. In contrast, neither the interaction between response

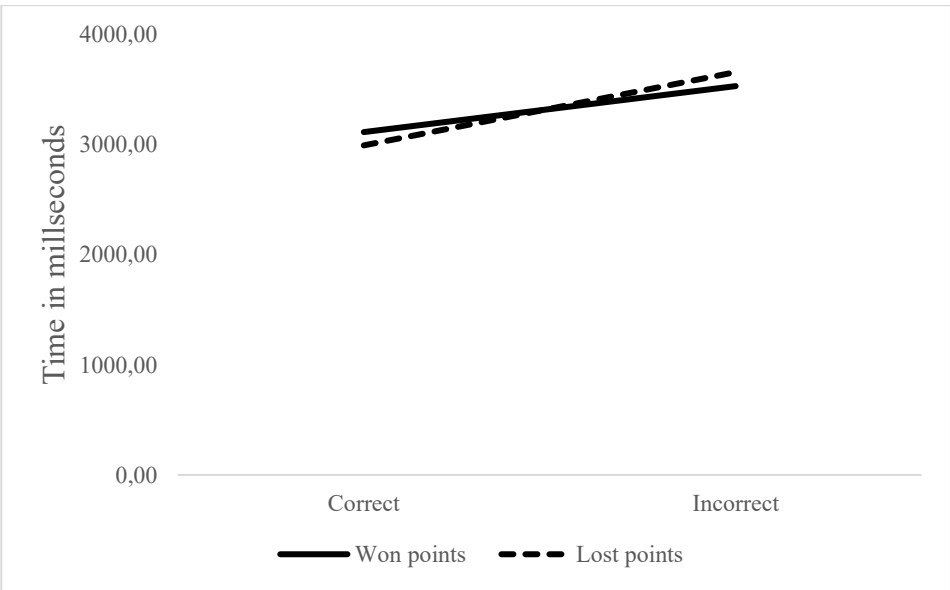
accuracy and gender, $F(1, 143) = 1.16$, $p = .284$, $\eta^2_p = .01$, nor the three-way interaction among response accuracy, type of situation, and gender was significant, $F(1, 143) = 0.43$, $p = .513$, $\eta^2_p < .01$. These findings indicate that gender did not moderate the effect of response accuracy on response time or the interaction between type of situation and response accuracy.

Table 2

Results of 2 x 2 x 2 mixed ANOVA

Independent variables	<i>df</i>	<i>F</i>	<i>p</i>	η^2_p
Response accuracy	1, 143	148.26	<.001	.51
Type of situation	1, 143	<.01	.992	<.01
Gender	1, 143	2.21	.140	.02
Response accuracy * type of situation	1, 143	7.23	.008	.05
Response accuracy * gender	1, 143	1.16	.284	.01
Gender * type of situation	1, 143	0.87	.341	.01
Response accuracy * gender * type of situation	1, 143	0.43	.513	<.01

Figure 1
Interaction of type of situation and response accuracy



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Discussion

In line with our hypothesis, the results showed faster response times for correct than for incorrect responses. On average, participants responded about half a second more quickly when they accurately recognized the nonverbal behavior of a player after a rally. Although our data do

not allow us to identify the precise mechanisms at work, these findings suggest that greater efficiency in decoding may be linked to higher accuracy, which could be consistent with the involvement of more automatic perceptual processes.

In line with relevant theoretical frameworks, such as the biocultural framework of nonverbal

behavior (Furley, 2021) and the dynamic interactive theory of person construal (Freeman & Ambady, 2011), decoding processes likely involve a complex interplay between more automatic bottom-up and higher order top-down mechanisms, with the relative contribution depending on the type of nonverbal behavior being decoded. While it is important to acknowledge that response time alone cannot directly assess the relative influence of these processes, the present findings provide tentative evidence that postperformance nonverbal behavior may be primarily decoded through intuitive, automatic processes. This interpretation is further supported by evolutionary perspectives on affective processing, which propose that

humans have evolved to rapidly and efficiently recognize affective expressions because such signals convey fitness-relevant information often visible in postperformance nonverbal behavior (e.g., Tracy & Robins, 2008). From an applied perspective, given the potential effects of observing postperformance nonverbal behavior (Fritsch, Redlich, et al., 2022), athletes may benefit from developing strategies that either direct attention toward or deliberately away from the behavior of an opponent or teammate (Beatty & Janelle, 2019). Notably, when situational constraints make it difficult to shift attention away, athletes might instead be advised to employ acceptance-based strategies, which may not change the perception of such behaviors but can shape the way they are interpreted (Birrer et al., 2012).

Exploratory analyses revealed that the difference in response times between correct and incorrect responses was more pronounced for videos showing players' nonverbal behavior after lost points. One explanation for this finding is the negativity bias stating that negative events have a stronger influence on individual's psychological state than positive ones (Baumeister et al., 2001). This bias may imply that particularly intense negative behaviors are more likely to be decoded efficiently. This interpretation is indirectly supported by a meta-analysis indicating that negative stimuli with higher arousal tend to elicit more intense processing than equally arousing positive stimuli, whereas no difference is observed between positive and negative stimuli

when arousal is low (Yuan et al., 2019). While previous meta-analytic evidence suggests that females are slightly more accurate than males in decoding affective cues (Hall et al., 2025), the present study did not reveal any gender effects on the association between response time and response accuracy. One possibility is that the specific stimulus material—postperformance nonverbal behavior in a competitive sports context—represents signals that are generally important to decode efficiently, thereby reducing the likelihood of gender-related differences emerging.

The negativity bias may further explain why lost points were easier to recognize than won points, which is consistent with findings from previous studies (Fritsch, Preine, Jekauc, 2022; Fritsch et al., 2023). The relevance of the negativity bias in relation to affective processes has been supported in a meta-analysis including 874 samples showing that procedures to induce negative affective states had effects nearly twice as large than those aiming to induce positive affect states (Joseph et al., 2020). Grounded in evolutionary accounts, it is assumed that the negativity bias is adaptive, enabling humans to show heightened awareness to negative information since it may signal a need to change one's behavior (Baumeister et al., 2001).

Limitations and Future Research

There are several limitations that should be addressed. First, although a study in tennis showed that tennis players' visible behavior is closely related to their emotional experience (Fritsch et al., 2024), it should be considered that the players' nonverbal behavior is not only associated with the players' underlying affective state but may also convey other cognitive states (Fridlund, 1994). Second, while a faster response time can be considered as an indicator of processing efficiency (Cardoso et al., 2021; Logan, 1988; Schneider & Shiffrin, 1977), there are also other factors that may influence response time, such as a higher certainty or more unambiguous footage (Elfenbein & Ambady, 2003). Nevertheless, in line with theoretical considerations (Furley,

2021), we interpret the results as supporting evidence that relying on one's intuitive impression when decoding nonverbal behavior is beneficial for accuracy. Third, we did not assess participants' previous experience with watching tennis, which might affect decoding accuracy, although previous research showed mixed effects in this regard (e.g., Fritsch, Preine, Jekauc, 2022; Furley & Schweizer, 2014).

Various methods may be used in future research to shed more light on the psychological processes behind the decoding of nonverbal behavior. For example, while we did not explicitly ask participants to respond as fast as possible to allow for natural response behavior, future studies could test this approach to examine the assumption that following one's automatic impression is beneficial for the accuracy of decoding nonverbal behavior. In addition, the use of eye tracking, combined with the consideration of domain-specific knowledge, could help decipher which body cues individuals focus on in the decoding process depending on their expertise. In the same vein, considering that the present study focused on post-performance nonverbal behavior, the comparison

of different types of nonverbal behavior (e.g., nonverbal behavior that does not directly follow a performance-relevant event) could provide further insights into the interplay between automatic perceptual and higher order cognitive processes in the decoding processes of nonverbal behavior (Furley, 2021).

Conclusions

The decoding of nonverbal behavior is a complex process that likely involves both higher order cognitive and more automatic perceptual components (Furley, 2021). The central finding of shorter response times for correct compared to incorrect responses provides first evidence that faster, possibly more intuitive impressions may be associated with greater accuracy in decoding post-performance nonverbal behavior. While the present design does not allow firm conclusions about the underlying mechanisms, these results highlight a promising direction for future research aimed at disentangling the relative contributions of more automatic and higher order cognitive decoding processes.

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During the preparation of this work, the authors used DeepL in order to improve readability of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed, and take full responsibility for the content of the publication.

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